

Ch28: 1, 17, 18, 30, 32, 46, 47

Basic score 4` + problem 17 2`+ problem 18 2`+ problem 47 2`

1. (a) We seek the electrostatic field established by the separation of charges (brought on by the magnetic force). With Eq. 28-10, we define the magnitude of the electric field as

$$|\vec{E}| = v|\vec{B}| = (20.0 \text{ m/s})(0.040 \text{ T}) = 0.800 \text{ V/m}.$$

Its direction may be inferred from Figure 28-8; its direction is opposite to that defined by  $\vec{v} \times \vec{B}$ . In summary,

$$\vec{E} = -(0.800 \text{ V/m})\hat{k}$$

which insures that  $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$  vanishes.

(b) Equation 28-9 yields  $V = Ed = (0.800 \text{ V/m})(2.00 \text{ m}) = 1.60 \text{ V}$ .

(c) The surface  $z = 0$

17. (a) The area of the loop is  $A = \frac{1}{2}(30\text{cm})(40\text{cm}) = 6.0 \times 10^2 \text{ cm}^2$ , so

$$\mu = iA = (7.5 \text{ A})(6.0 \times 10^{-2} \text{ m}^2) = 0.45 \text{ A} \cdot \text{m}^2. \quad 1 \text{ point} \quad \text{公式结果各 } 0.5 \text{ 分}$$

(b) The torque on the loop is

$$\tau = \mu B \sin \theta = (0.45 \text{ A} \cdot \text{m}^2)(120 \times 10^3 \text{ T}) \sin 90^\circ = 5.4 \times 10^{-2} \text{ N} \cdot \text{m}.$$

1 point 公式结果各 0.5 分

18. (a) The accelerating process may be seen as a conversion of potential energy  $eV$  into kinetic energy. Since it starts from rest,  $\frac{1}{2}m_e v^2 = eV$  and

$$v = \sqrt{\frac{2eV}{m_e}} = \sqrt{\frac{2(1.60 \times 10^{-19} \text{ C})(380 \text{ V})}{9.11 \times 10^{-31} \text{ kg}}} = 1.16 \times 10^7 \text{ m/s.} \quad \text{1 point 公式结果各 0.5 分}$$

(b) Equation 28-16 gives

$$r = \frac{m_e v}{eB} = \frac{(9.11 \times 10^{-31} \text{ kg})(1.16 \times 10^7 \text{ m/s})}{(1.60 \times 10^{-19} \text{ C})(200 \times 10^{-3} \text{ T})} = 3.30 \times 10^{-4} \text{ m.} \quad \text{1 point 公式结果各 0.5 分}$$

30. We use  $d\vec{F}_B = id\vec{L} \times \vec{B}$ , where  $d\vec{L} = -dx\hat{i}$  and  $\vec{B} = B_x\hat{i} + B_y\hat{j}$ . Thus,

$$\begin{aligned} \vec{F}_B &= \int_{\text{wire}} id\vec{L} \times \vec{B} = \int_{1.0}^{3.0} idx\hat{i} \times (B_x\hat{i} + B_y\hat{j}) = \int_{1.0}^{3.0} iB_y dx \hat{k} \\ &= (7.0) \int_{1.0}^{3.0} 8.0x^2 dx \hat{k} = -0.49\hat{k} \text{ N} \end{aligned}$$

32. (a) The magnetic force on the wire is  $F_B = idB$ , pointing to the left. Thus

$$\begin{aligned} v = at &= \frac{F_B t}{m} = \frac{idBt}{m} = \frac{(9.13 \times 10^{-3} \text{ A})(2.56 \times 10^{-2} \text{ m})(7.35 \times 10^{-2} \text{ T})(0.0611 \text{ s})}{2.41 \times 10^{-5} \text{ kg}} \\ &= 4.35 \times 10^{-2} \text{ m/s.} \end{aligned}$$

(b) The direction is to the left (away from the generator).

46. (a) The kinetic energy gained is due to the potential energy decrease as the dipole swings from a position specified by angle  $\theta$  to that of being aligned (zero angle) with the field. Thus,

$$K = U_i - U_f = -\mu B \cos \theta - (-\mu B \cos 0^\circ)$$

Therefore, using SI units, the angle is

$$\theta = \cos^{-1} \left( 1 - \frac{K}{\mu B} \right) = \cos^{-1} \left( 1 - \frac{0.00080}{(0.020)(0.046)} \right) = 83^\circ.$$

(b) Since we are making the assumption that no energy is dissipated in this process, then the dipole will continue its rotation (similar to a pendulum) until it reaches an angle  $\theta = 83^\circ$  on the other side of the alignment axis.

**47.** When the electric and magnetic forces are in balance

$$eE = ev_d B \Rightarrow v_d = \frac{E}{B} = \frac{V}{Bd} \quad 0.5 \text{ point}$$

The drift speed in the circuit is:

$$v_d = \frac{J}{ne} = \frac{i}{Ane} = \frac{i}{(ld)ne} \quad 0.5 \text{ point}$$

$$v_d = \frac{i}{(ld)ne} = \frac{V}{Bd} \Rightarrow V = \frac{iB}{nle} \quad 0.5 \text{ point}$$

$$= \frac{(57\text{A})(0.65\text{ T})}{(8.47 \times 10^{28}/\text{m}^3)(75\mu\text{m})(1.6 \times 10^{-19}\text{C})} = 3.6 \times 10^{-5}\text{V}. \quad 0.5 \text{ point}$$

(直接写最后一个公式，可不扣分)